OpenLearn | Free learning from The Open University

The Open University

Statement of participation

Hemant Kumar Sah

has completed the free course including any mandatory tests for:

Introduction to complex analysis

This 12-hour course explored differentiation and integration of complex functions.

Issue date: 3 May 2026



www.open.edu/openlearn

This statement does not imply the award of credit points nor the conferment of a University Qualification.
This statement confirms that this free course and all mandatory tests were passed by the learner.

Please go to the course on OpenLearn for full details:
https://www.open.edu/openlearn/science-maths-technology/introduction-complex-analysis/content-section-overview

COURSE CODE: **M337_2**

OpenLearn | Free learning from The Open University

The Open University

Introduction to complex analysis

https://www.open.edu/openlearn/science-maths-technology/introduction-complex-analysis/content-section-overview

Course summary

This OpenLearn course is an extract from the third level Open University course M337 Complex Analysis, which develops the theory of functions of a complex variable, emphasising their geometric properties and indicating some applications.

Learning outcomes

By completing this course, the learner should be able to:

- use the definition of derivative to show that a given function is or is not differentiable at a point
- use the Cauchy–Riemann equations to show that a function is or is not differentiable at a point
- interpret the derivative of a complex function at a point as a rotation and a scaling of a small disc
- appreciate how complex integrals can be defined by analogy with real integrals
- define the integral of a complex function along a contour and evaluate such integrals
- state and use several key theorems to evaluate contour integrals.

Completed study

The learner has completed the following:

Session 1

Session 1: Differentiation

Session 2

Session 2: Integration